CASE 07 — GOVERNANCE

CIVICTRACE WORKING PROPOSAL

“One Complaint. Five Departments.”

Making complaint ownership and handling history easy to trace across departmental handoffs.



Team: Synergy Trinity

Members: Jhnavi Bhadkariya (TL), Lavanya Bajpai, Aaryanvi Sharma

Speaker Track (25s): Welcome, judges. Our proposal addresses Case 07: Governance. When public grievances cross municipal boundaries, clear ownership can become obscured. CivicTrace is a human-supervised workflow overlay designed to make complaint handoffs, decisions, and ownership completely traceable.

Multi-Channel Intake & Municipal Silos

CASE UNDERSTANDING

Grievances flow from fragmented public touchpoints into siloed departmental workflows.

Intake Channels & Stakeholders

WEB PORTAL | CALL CENTER | SOCIAL MEDIA | IN-PERSON DESK

Core Stakeholders:

- **Residents:** Submit complaints, seek accountable status updates
- **Intake Staff:** Log initial entries and route to municipal units
- **Department Handlers:** Execute field repairs and record updates
- **Ward Reviewers:** Oversee inter-departmental case transfers

Case Signals vs. Unknowns

SIMULATED CASE SIGNAL

Case brief cites a 23% reassignment rate across unlinked department systems.

KNOWN LIMITATION / UNKNOWN

Reassignment alone does not prove incorrect routing or resolution delays in a given city.

Speaker Track (25s): Public complaints enter municipal systems through multiple touchpoints but soon hit isolated departmental systems. While the case scenario cites a 23% reassignment rate, we explicitly recognize this as simulated prompt data rather than a verified statistic for any real municipality.

Evaluation of Problem Framings

STRATEGIC DECISION

Comparing candidate hypotheses to select the most testable framing for empirical investigation.

OPTION A

Routing Delays

Hypothesis: Reassignments drive resolution delays.

Unknown: Reassignment does not inherently prove incorrect routing, avoidable transfer, or delay.

OPTION B — SELECTED

PROVISIONAL

Fragmented Ownership

Focus: Unclear ownership & untraceable case histories across handoffs.

Reason: Most testable hypothesis to empirically validate and solve.

OPTION C

Safety Auditability

Focus: High-consequence safety risk prioritisation.

Unknown: Case material does not establish actual safety failure evidence.

Data Clarification: The 23% reassignment statistic in the case scenario is simulated competition material, not an external real-world statistic. Option B remains a provisional choice subject to empirical confirmation.

Speaker Track (25s): We evaluated three candidate framings. Option B—fragmented complaint ownership—was chosen provisionally because missing ownership history is empirically testable, whereas claiming proven delays or safety catastrophes would exceed the case evidence.

Provisional Problem Statement

HYPOTHESIS UNDER VALIDATION

> **"When complaints arrive through multiple channels or move between departments, the workflow may not consistently make the accountable office, assignment history, status changes and transfer reasons easy to trace.**
>
> This may make it difficult for residents and authorised reviewers to determine who is responsible for the next action and how the complaint was handled."

What Investigation Must Establish

1. Do departmental handoffs cause unclear ownership?
2. Is decision history missing during transfers?
3. Do unrecorded reassignments add measurable resolution time?

Evidence Boundary Constraint

This gap must be validated empirically in local systems before claiming that it causes systematic delay, harm, or administrative failure in any named municipality.

Speaker Track (25s): Our problem statement defines specific, measurable targets for field investigation. It asks whether departmental handoffs create unclear ownership or missing records, without asserting that these failures are established facts in any real-world city.

CivicTrace Solution Concept

HUMAN-SUPERVISED WORKSPACE

A proposed complaint traceability workspace overlay — designed to complement, not replace, municipal systems.

Persistent Case ID

Unified tracking reference generated at intake across all channels.

Responsible Office

Explicit current assigned unit and readable status indicator.

Handoff Timeline

Chronological action log recording all inter-office transfers.

Recorded Transfer Reason

Mandatory reason field required whenever a case is reassigned.

System Scope & Boundaries

- **Traceability Layer:** Sits as a lightweight audit workspace above existing software.
- **Complements Current ERPs:** Does not require replacing working municipal databases.
- **Human Oversight:** Department officials confirm all transfers, reassignments, and closures.

Speaker Track (25s): CivicTrace is proposed as a human-supervised traceability layer, not a replacement for existing municipal software. It provides four core capabilities: persistent IDs, explicit responsible offices, chronological handoff histories, and mandatory reassignment reasons.

Workflow & AI/Human Safeguards

OPERATIONAL ARCHITECTURE

Clear boundary between automated assistance and human operational control.

1. Complaint Intake
Original text preserved verbatim

2. AI Suggestion
Draft category & department tag

3. Human Confirmation
Officer reviews & confirms transfer

4. Timeline Appended
Auditable transfer log recorded

Strict Automation Safeguards & Governance

NO Auto-Rejection / Closure: AI can never dismiss or close complaints.

"Needs Human Review" Pathway: Direct human routing for low confidence or safety risks.

NO Silent Merging: Duplicate suggestions require explicit human approval.

Access & Audit Logs: Role-based access controls and complete transfer auditability.

Speaker Track (25s): Our workflow enforces strict human oversight. AI provides suggestions for categories and duplicate flags, but human officers must approve every reassignment. The system cannot auto-reject, auto-close, or silently merge complaints.

Proposed Prototype Testing Plan

TARGET ACCEPTANCE CRITERIA

Implementation Status Note: The software prototype has not yet been built. The criteria below represent proposed target benchmarks for future testing on synthetic datasets by our 3-person CSE team.

PROPOSED TEST I: PERSISTENCE

ID Stability Benchmark

Target: 30 / 30 synthetic test cases retain stable ID and complete handoff history across transfers.

PROPOSED TEST II: RECONSTRUCTABILITY

Timeline Audit Benchmark

Target: Independent reviewers correctly reconstruct ≥90% of 20 synthetic case handling histories.

PROPOSED TEST III: SAFETY GUARD

Zero Auto-Closure Test

Target: Zero AI-only closures; 100% of seeded safety cases flagged for mandatory human review.

Proposed Evaluation Metrics: Baseline timeline reconstruction accuracy vs. post-prototype reconstruction accuracy; missed safety review rates evaluated across intake languages.

Speaker Track (25s): Designed for a 3-person CSE team, our testing plan establishes proposed acceptance benchmarks using synthetic cases. These focus on verifying ID stability, history reconstructability, and zero AI auto-closures prior to any potential pilot.

Risks, Build Roadmap & Source Scope

ROUND 2 ROADMAP

Primary Operational Risks

- **False Duplicate Links:** Mitigated by mandatory human confirmation.
- **Missing Transfer Logs:** Mitigated by mandatory reason fields.
- **Language Exclusion:** Mitigated by preserving raw multilingual text.

Round 2 Build Roadmap

1. Core timeline & manual transfer UI | 2. Add AI department suggestions & 'Needs Human Review' fallback | 3. Synthetic dataset testing & failure logging.

Verified Source Scope & Limitations

- **Aggarwal et al. (2010):** *Int. J. Public Sector Perform. Mgt.* DOI: 10.3233/IJR-120095 — Context baseline; older, few-city sample.
- **DARPG Guidelines (2024):** Dept. of Administrative Reforms & Public Grievances, GoI [PDF Link] — Central CPGRAMS framework; not local municipal proof.
- **CAG UP Local Bodies Report No. 2 (2015):** Comptroller & Auditor General of India [PDF Link] — Historical audit; specific selected ULBs.
- **ACM DIS '26 Companion:** DOI: 10.1145/3802974.3810897 — Extended abstract; sample limitations.

CORE VALIDATION QUESTION:
"Can a lightweight ownership timeline improve history reconstructability without adding administrative overhead?"

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Speaker Track (25s): We acknowledge key operational risks like false duplicate links and present our Round 2 build roadmap. We explicitly cite our external references with their scope and limitations, framing central guidelines and audits as context rather than direct proof of municipal failure.